Project Allotment: EC-101 [B. TECH SEM 1]

AI, Inverse AI Regression and Optimization for Antenna Design Dataset

Dataset structure: DV1–DV10 are the 10 design variables. F1–F5 are five frequencies. P1–P7 are seven performance indicators evaluated at each frequency, producing 35 performance outputs (P1_F1 … P7_F5).

1. Common Problem Definition

Forward regression: [DV1 … DV10] → [P1_F1 … P7_F5] (10 inputs → 35 outputs).

Inverse design: Target [P1_F1 … P7_F5] → candidate [DV1 … DV10], preferably validated by a forward surrogate model and, where possible, CST/experimental evaluation.

DV1, DV2 and DV3 are responsible for F1, F2,..,F5, (P3, P4) at F1, (P3, P4) at F2,.., (P3, P4); DV4 is responsible for (P1, P2) at F1, (P1, P2) at F2,..., (P1, P2) at F5, DV5 to DV10 are responsible for (P5, P6, P7, P8, P9, P10)

All students should use a common dataset version, preprocessing protocol, train/validation/test split, and reporting metrics so that algorithms can be compared fairly.

2. Common Deliverables for Every Student

- Clean, well-documented source code.
- README with installation and execution instructions.
- Dataset preprocessing and normalization script.
- Training and testing script.
- Prediction/inverse-design script.
- Performance metrics: MAE, RMSE, R^2 and prediction time; inverse projects should additionally report target-versus-achieved performance error.
- Plots appropriate to the algorithm, including predicted-versus-actual results and/or convergence curves.
- Short project report describing algorithm, methodology, results, limitations and future work.

3. Individual Project Allotment

Roll No.	Project / Algorithm	Type	Input	Output	Required Evaluation	Main Deliverable	Email ID
26BCS001	Multiple Linear Regression	Forward	DV1–DV10	35 performance outputs	MAE, RMSE, R^2; baseline comparison	Documented Python code + results + report	26BCS001@nith.ac.in
26BCS002	Polynomial Regression	Forward	DV1–DV10	35 performance outputs	MAE, RMSE, R^2; polynomial degree study	Documented Python code + results + report	26BCS002@nith.ac.in
26BCS003	Ridge Regression	Forward	DV1–DV10	35 performance outputs	MAE, RMSE, R^2; regularization tuning	Documented Python code + results + report	26BCS003@nith.ac.in
26BCS004	Lasso Regression	Forward	DV1–DV10	35 performance outputs	MAE, RMSE, R^2; sparsity analysis	Documented Python code + results + report	26BCS004@nith.ac.in
26BCS005	Elastic Net Regression	Forward	DV1–DV10	35 performance outputs	MAE, RMSE, R^2	Documented Python code + results + report	26BCS005@nith.ac.in
26BCS006	K-Nearest Neighbours Regression	Forward	DV1–DV10	35 performance outputs	MAE, RMSE, R^2; k sensitivity	Documented Python code + results + report	26BCS006@nith.ac.in
26BCS007	Support Vector Regression	Forward	DV1–DV10	35 performance outputs	MAE, RMSE, R^2; kernel comparison	Documented Python code + results + report	26BCS007@nith.ac.in
26BCS008	Decision Tree Regression	Forward	DV1–DV10	35 performance outputs	MAE, RMSE, R^2; tree complexity	Documented Python code + results + report	26BCS008@nith.ac.in

26BCS009	Random Forest Regression	Forward	DV1–DV10	35 performance outputs	MAE, RMSE, R^2; feature importance	Documented Python code + results + report	26BCS009@nith.ac.in
26BCS010	Extra Trees Regression	Forward	DV1–DV10	35 performance outputs	MAE, RMSE, R^2	Documented Python code + results + report	26BCS010@nith.ac.in
26BCS011	Gradient Boosting Regression	Forward	DV1–DV10	35 performance outputs	MAE, RMSE, R^2	Documented Python code + results + report	26BCS011@nith.ac.in
26BCS012	AdaBoost Regression	Forward	DV1–DV10	35 performance outputs	MAE, RMSE, R^2	Documented Python code + results + report	26BCS012@nith.ac.in
26BCS013	Histogram Gradient Boosting Regression	Forward	DV1–DV10	35 performance outputs	MAE, RMSE, R^2	Documented Python code + results + report	26BCS013@nith.ac.in
26BCS014	Gaussian Process Regression	Forward	DV1–DV10	35 performance outputs	MAE, RMSE, R^2; uncertainty	Documented Python code + results + report	26BCS014@nith.ac.in
26BCS015	Bayesian Ridge Regression	Forward	DV1–DV10	35 performance outputs	MAE, RMSE, R^2	Documented Python code + results + report	26BCS015@nith.ac.in
26BCS016	Kernel Ridge Regression	Forward	DV1–DV10	35 performance outputs	MAE, RMSE, R^2	Documented Python code + results + report	26BCS016@nith.ac.in
26BCS017	Partial Least Squares Regression	Forward	DV1–DV10	35 performance outputs	MAE, RMSE, R^2; component study	Documented Python code + results + report	26BCS017@nith.ac.in
26BCS018	Principal Component Regression	Forward	DV1–DV10	35 performance outputs	MAE, RMSE, R^2	Documented Python code + results + report	26BCS018@nith.ac.in
26BCS019	Huber Regression	Forward	DV1–DV10	35 performance outputs	MAE, RMSE, R^2; robustness	Documented Python code + results + report	26BCS019@nith.ac.in
26BCS020	Quantile Regression	Forward	DV1–DV10	35 performance outputs	Quantile loss; uncertainty intervals	Documented Python code + results + report	26BCS020@nith.ac.in
26BCS021	Genetic Algorithm	Inverse	Target 35 performance values	DV1–DV10	Inverse error; convergence; achieved performance	Documented Python code + results + report	26BCS021@nith.ac.in
26BCS022	Particle Swarm Optimization	Inverse	Target 35 performance values	DV1–DV10	Inverse error; convergence	Documented Python code + results + report	26BCS022@nith.ac.in
26BCS023	Differential Evolution	Inverse	Target 35 performance values	DV1–DV10	Inverse error; convergence	Documented Python code + results + report	26BCS023@nith.ac.in
26BCS024	Artificial Bee Colony	Inverse	Target 35 performance values	DV1–DV10	Inverse error; convergence	Documented Python code + results + report	26BCS024@nith.ac.in
26BCS025	Ant Colony Optimization	Inverse	Target 35 performance values	DV1–DV10	Inverse error; convergence	Documented Python code + results + report	26BCS025@nith.ac.in
26BCS026	Grey Wolf Optimizer	Inverse	Target 35 performance values	DV1–DV10	Inverse error; convergence	Documented Python code + results + report	26BCS026@nith.ac.in
26BCS027	Whale Optimization Algorithm	Inverse	Target 35 performance values	DV1–DV10	Inverse error; convergence	Documented Python code + results + report	26BCS027@nith.ac.in
26BCS028	Firefly Algorithm	Inverse	Target 35 performance values	DV1–DV10	Inverse error; convergence	Documented Python code + results + report	26BCS028@nith.ac.in
26BCS029	Coyote Optimization Algorithm	Inverse	Target 35 performance values	DV1–DV10	Inverse error; convergence	Documented Python code + results + report	26BCS029@nith.ac.in
26BCS030	Harris Hawks Optimization	Inverse	Target 35 performance values	DV1–DV10	Inverse error; convergence	Documented Python code + results + report	26BCS030@nith.ac.in
26BCS031	Salp Swarm Algorithm	Inverse	Target 35 performance values	DV1–DV10	Inverse error; convergence	Documented Python code + results + report	26BCS031@nith.ac.in
26BCS032	Moth-Flame Optimization	Inverse	Target 35 performance values	DV1–DV10	Inverse error; convergence	Documented Python code + results + report	26BCS032@nith.ac.in
26BCS033	Bat Algorithm	Inverse	Target 35 performance values	DV1–DV10	Inverse error; convergence	Documented Python code + results + report	26BCS033@nith.ac.in
26BCS034	Flower Pollination Algorithm	Inverse	Target 35 performance values	DV1–DV10	Inverse error; convergence	Documented Python code + results + report	26BCS034@nith.ac.in
26BCS035	Teaching–Learning-Based Optimization	Inverse	Target 35 performance values	DV1–DV10	Inverse error; convergence	Documented Python code + results + report	26BCS035@nith.ac.in
26BCS036	Harmony Search	Inverse	Target 35 performance values	DV1–DV10	Inverse error; convergence	Documented Python code + results + report	26BCS036@nith.ac.in
26BCS037	Sine Cosine Algorithm	Inverse	Target 35 performance values	DV1–DV10	Inverse error; convergence	Documented Python code + results + report	26BCS037@nith.ac.in
26BCS038	Sparrow Search Algorithm	Inverse	Target 35 performance values	DV1–DV10	Inverse error; convergence	Documented Python code + results + report	26BCS038@nith.ac.in

26BCS039	Marine Predators Algorithm	Inverse	Target 35 performance values	DV1–DV10	Inverse error; convergence	Documented Python code + results + report	26BCS039@nith.ac.in
26BCS040	Dragonfly Algorithm	Inverse	Target 35 performance values	DV1–DV10	Inverse error; convergence	Documented Python code + results + report	26BCS040@nith.ac.in
26BCS041	ANN + Particle Swarm Optimization	Hybrid Inverse	Target performance + DV bounds	DV1–DV10	Surrogate accuracy + inverse error	Documented Python code + results + report	26BCS041@nith.ac.in
26BCS042	ANN + Genetic Algorithm	Hybrid Inverse	Target performance + DV bounds	DV1–DV10	Surrogate accuracy + inverse error	Documented Python code + results + report	26BCS042@nith.ac.in
26BCS043	ANN + Differential Evolution	Hybrid Inverse	Target performance + DV bounds	DV1–DV10	Surrogate accuracy + inverse error	Documented Python code + results + report	26BCS043@nith.ac.in
26BCS044	ANN + Grey Wolf Optimizer	Hybrid Inverse	Target performance + DV bounds	DV1–DV10	Surrogate accuracy + inverse error	Documented Python code + results + report	26BCS044@nith.ac.in
26BCS045	ANN + Whale Optimization Algorithm	Hybrid Inverse	Target performance + DV bounds	DV1–DV10	Surrogate accuracy + inverse error	Documented Python code + results + report	26BCS045@nith.ac.in
26BCS046	ANN + Coyote Optimization Algorithm	Hybrid Inverse	Target performance + DV bounds	DV1–DV10	Surrogate accuracy + inverse error	Documented Python code + results + report	26BCS046@nith.ac.in
26BCS047	Random Forest + PSO	Hybrid Inverse	Target performance + DV bounds	DV1–DV10	Surrogate accuracy + inverse error	Documented Python code + results + report	26BCS047@nith.ac.in
26BCS048	Random Forest + GA	Hybrid Inverse	Target performance + DV bounds	DV1–DV10	Surrogate accuracy + inverse error	Documented Python code + results + report	26BCS048@nith.ac.in
26BCS049	SVR + PSO	Hybrid Inverse	Target performance + DV bounds	DV1–DV10	Surrogate accuracy + inverse error	Documented Python code + results + report	26BCS049@nith.ac.in
26BCS050	SVR + GA	Hybrid Inverse	Target performance + DV bounds	DV1–DV10	Surrogate accuracy + inverse error	Documented Python code + results + report	26BCS050@nith.ac.in
26BCS051	Gaussian Process + Bayesian Optimization	Hybrid Inverse	Target performance + DV bounds	DV1–DV10	Uncertainty + inverse error	Documented Python code + results + report	26BCS051@nith.ac.in
26BCS052	Gaussian Process + PSO	Hybrid Inverse	Target performance + DV bounds	DV1–DV10	Surrogate accuracy + inverse error	Documented Python code + results + report	26BCS052@nith.ac.in
26BCS053	KNN + Genetic Algorithm	Hybrid Inverse	Target performance + DV bounds	DV1–DV10	Surrogate accuracy + inverse error	Documented Python code + results + report	26BCS053@nith.ac.in
26BCS054	Extra Trees + PSO	Hybrid Inverse	Target performance + DV bounds	DV1–DV10	Surrogate accuracy + inverse error	Documented Python code + results + report	26BCS054@nith.ac.in
26BCS055	Gradient Boosting + PSO	Hybrid Inverse	Target performance + DV bounds	DV1–DV10	Surrogate accuracy + inverse error	Documented Python code + results + report	26BCS055@nith.ac.in
26BCS056	Polynomial Regression + PSO	Hybrid Inverse	Target performance + DV bounds	DV1–DV10	Surrogate accuracy + inverse error	Documented Python code + results + report	26BCS056@nith.ac.in
26BCS057	Ridge Regression + Differential Evolution	Hybrid Inverse	Target performance + DV bounds	DV1–DV10	Surrogate accuracy + inverse error	Documented Python code + results + report	26BCS057@nith.ac.in
26BCS058	Kernel Ridge + Genetic Algorithm	Hybrid Inverse	Target performance + DV bounds	DV1–DV10	Surrogate accuracy + inverse error	Documented Python code + results + report	26BCS058@nith.ac.in
26BCS059	Ensemble Regression + PSO	Hybrid Inverse	Target performance + DV bounds	DV1–DV10	Surrogate accuracy + inverse error	Documented Python code + results + report	26BCS059@nith.ac.in
26BCS060	Ensemble Regression + Multi-Objective GA	Hybrid Inverse	Target performance + DV bounds	DV1–DV10	Pareto solutions + inverse error	Documented Python code + results + report	26BCS060@nith.ac.in
26BCS061	Voting Regressor	Multi-Classical	DV1–DV10	35 performance outputs	MAE, RMSE, R^2	Documented Python code + results + report	26BCS061@nith.ac.in
26BCS062	Weighted Voting Regression	Multi-Classical	DV1–DV10	35 performance outputs	MAE, RMSE, R^2; weight optimization	Documented Python code + results + report	26BCS062@nith.ac.in
26BCS063	Stacking Regression	Multi-Classical	DV1–DV10	35 performance outputs	MAE, RMSE, R^2; meta-model study	Documented Python code + results + report	26BCS063@nith.ac.in
26BCS064	Blending Regression	Multi-Classical	DV1–DV10	35 performance outputs	MAE, RMSE, R^2	Documented Python code + results + report	26BCS064@nith.ac.in
26BCS065	Random Forest + Extra Trees Ensemble	Multi-Classical	DV1–DV10	35 performance outputs	MAE, RMSE, R^2	Documented Python code + results + report	26BCS065@nith.ac.in
26BCS066	SVR + Random Forest Ensemble	Multi-Classical	DV1–DV10	35 performance outputs	MAE, RMSE, R^2	Documented Python code + results + report	26BCS066@nith.ac.in
26BCS067	KNN + SVR + Random Forest Ensemble	Multi-Classical	DV1–DV10	35 performance outputs	MAE, RMSE, R^2	Documented Python code + results + report	26BCS067@nith.ac.in
26BCS068	Linear + Polynomial + Kernel Regression Ensemble	Multi-Classical	DV1–DV10	35 performance outputs	MAE, RMSE, R^2	Documented Python code + results + report	26BCS068@nith.ac.in

26BCS069	PCA + Multiple Regressor Ensemble	Multi-Classical	DV1–DV10	35 performance outputs	MAE, RMSE, R^2; PCA components	Documented Python code + results + report	26BCS069@nith.ac.in
26BCS070	Dynamic Model Selection Ensemble	Multi-Classical	DV1–DV10	35 performance outputs	MAE, RMSE, R^2; local model selection	Documented Python code + results + report	26BCS070@nith.ac.in
26BCS071	GA + PSO	Multi-Nature-Inspired	Target performance + DV bounds	DV1–DV10	Inverse error; convergence	Documented Python code + results + report	26BCS071@nith.ac.in
26BCS072	PSO + Differential Evolution	Multi-Nature-Inspired	Target performance + DV bounds	DV1–DV10	Inverse error; convergence	Documented Python code + results + report	26BCS072@nith.ac.in
26BCS073	PSO + Grey Wolf Optimizer	Multi-Nature-Inspired	Target performance + DV bounds	DV1–DV10	Inverse error; convergence	Documented Python code + results + report	26BCS073@nith.ac.in
26BCS074	GA + Grey Wolf Optimizer	Multi-Nature-Inspired	Target performance + DV bounds	DV1–DV10	Inverse error; convergence	Documented Python code + results + report	26BCS074@nith.ac.in
26BCS075	GA + Whale Optimization Algorithm	Multi-Nature-Inspired	Target performance + DV bounds	DV1–DV10	Inverse error; convergence	Documented Python code + results + report	26BCS075@nith.ac.in
26BCS076	Differential Evolution + PSO	Multi-Nature-Inspired	Target performance + DV bounds	DV1–DV10	Inverse error; convergence	Documented Python code + results + report	26BCS076@nith.ac.in
26BCS077	Coyote Optimization + PSO	Multi-Nature-Inspired	Target performance + DV bounds	DV1–DV10	Inverse error; convergence	Documented Python code + results + report	26BCS077@nith.ac.in
26BCS078	Grey Wolf + Whale Optimization	Multi-Nature-Inspired	Target performance + DV bounds	DV1–DV10	Inverse error; convergence	Documented Python code + results + report	26BCS078@nith.ac.in
26BCS079	Multi-Stage Nature-Inspired Optimization	Multi-Nature-Inspired	Target performance + DV bounds	DV1–DV10	Inverse error; convergence; stage comparison	Documented Python code + results + report	26BCS079@nith.ac.in
26BCS080	Adaptive Multi-Optimizer	Multi-Nature-Inspired	Target performance + DV bounds	DV1–DV10	Inverse error; convergence; switching strategy	Documented Python code + results + report	26BCS080@nith.ac.in
26BCS081	Direct Multi-Output ANN	Forward AI	DV1–DV10	35 performance outputs	MAE, RMSE, R^2	Documented Python code + results + report	26BCS081@nith.ac.in
26BCS082	Separate ANN per Performance Indicator	Forward AI	DV1–DV10	35 performance outputs grouped by P1–P7	MAE, RMSE, R^2	Documented Python code + results + report	26BCS082@nith.ac.in
26BCS083	Separate ANN per Frequency	Forward AI	DV1–DV10	35 performance outputs grouped by F1–F5	MAE, RMSE, R^2	Documented Python code + results + report	26BCS083@nith.ac.in
26BCS084	Direct Inverse ANN	Inverse AI	35 performance values	DV1–DV10	DV error + achieved performance error	Documented Python code + results + report	26BCS084@nith.ac.in
26BCS085	Forward ANN + Inverse ANN Consistency System	Forward/Inverse AI	DV ↔ performance	DV1–DV10 and 35 outputs	Round-trip consistency + inverse error	Documented Python code + results + report	26BCS085@nith.ac.in
26BCS086	Autoencoder Performance Compression + Regression	Forward AI	DV1–DV10	Compressed/reconstructed 35 outputs	Reconstruction error + regression error	Documented Python code + results + report	26BCS086@nith.ac.in
26BCS087	PCA + ANN Surrogate	Forward AI	DV1–DV10	35 performance outputs via PCA reconstruction	MAE, RMSE, R^2 + reconstruction	Documented Python code + results + report	26BCS087@nith.ac.in
26BCS088	Extreme Learning Machine	Forward AI	DV1–DV10	35 performance outputs	MAE, RMSE, R^2 + training time	Documented Python code + results + report	26BCS088@nith.ac.in
26BCS089	Radial Basis Function Neural Network	Forward AI	DV1–DV10	35 performance outputs	MAE, RMSE, R^2	Documented Python code + results + report	26BCS089@nith.ac.in
26BCS090	Multi-Objective Inverse Design System	Advanced Inverse	Selected target performance constraints	Multiple candidate DV1–DV10 designs	Pareto front + target-achievement error	Documented Python code + results + report	26BCS090@nith.ac.in

4. Recommended Validation Protocol

- Use identical train/validation/test splits for all projects.
- Fit preprocessing only on training data and apply the learned transformation to validation/test data.
- Do not select the best algorithm using the test set; reserve it for final comparison.
- For inverse projects, evaluate the proposed design by passing DV1–DV10 through a validated forward surrogate. If CST validation is available later, report surrogate-versus-CST differences separately.
- Because inverse antenna design can be many-to-one, do not judge an inverse method only by error in DV values; prioritize error between requested and achieved performance.
- Report random seed, hyperparameters, computational time and software/library versions.

5. Final Integrated Outcome

The semester can produce a common AI-for-Antenna-Design benchmark consisting of classical forward regression models, direct inverse AI models, nature-inspired inverse optimizers, hybrid surrogate–optimizer systems, multi-classical ensembles and multi-nature-inspired algorithms. The strongest codes can subsequently be integrated into a common performance-to-design-variable antenna design platform.